

BLAGOVESHCHENSK, Russian Federation

WMO: 315100

Lat: **50.280N**Lon: **127.478E**Elev: **555**StdP: **14.40**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-21.9	-31.2	1.0	-25.1	-27.4	1.2	-20.6	14.1	2.8	12.3	1.6	2.5	260	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	88.2	70.0	85.0	69.0	82.2	68.1	74.2	82.6	72.5	80.5	70.8	78.7	5.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	119.7	78.9	69.8	112.1	76.9	68.2	105.8	75.0	38.3	82.6	36.7	80.6	35.2	79.0	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.0	10.4	DB	-28.6	93.8	5.3	3.3	-32.4	96.1	-35.5	98.0	-38.4	99.9	-42.3	102.2	(4)
(5)			WB	-29.0	77.7	5.2	2.6	-32.7	79.6	-35.8	81.1	-38.7	82.6	-42.4	84.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	-4.9	3.5	20.4	40.8	55.7	67.5	72.5	68.3	56.1	38.8	15.3	-2.7	(6)	
	DBStd	28.71	8.81	8.94	10.72	7.78	7.38	6.36	4.52	5.29	7.06	8.48	11.57	8.14	(7)	
	HDD50	7297	1702	1303	917	292	27	0	0	0	24	358	1041	1633	(8)	
	HDD65	11055	2167	1723	1382	726	302	46	4	27	277	812	1491	2098	(9)	
	CDD50	2227	0	0	0	16	204	524	698	568	206	11	0	0	(10)	
	CDD65	508	0	0	0	0	14	119	237	129	9	0	0	0	(11)	
	CDH74	3967	0	0	0	14	243	1082	1706	824	96	2	0	0	(12)	
	CDH80	1133	0	0	0	2	65	359	509	189	9	0	0	0	(13)	
(14) Wind	WSAvg	4.5	4.0	4.2	5.2	5.8	5.4	4.3	3.9	4.0	4.3	4.9	4.4	4.0	(14)	
(15) Precipitation	PrecAvg	22.0	0.2	0.2	0.4	1.2	1.9	3.3	5.3	4.8	2.8	0.9	0.5	0.4	(15)	
	PrecMax	31.9	1.4	0.8	1.5	5.5	5.6	7.5	10.6	9.9	7.1	3.3	1.6	0.9	(16)	
	PrecMin	14.1	0.0	0.0	0.0	0.1	0.2	0.2	0.9	0.0	0.4	0.0	0.0	0.1	(17)	
	PrecStd	3.7	0.2	0.2	0.4	0.9	1.2	1.9	2.4	2.2	1.6	0.7	0.4	0.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.0	34.1	52.7	75.3	86.6	93.1	92.9	88.5	81.0	68.3	47.8	22.7	(19)	
		MCWB	16.6	27.1	39.1	52.9	59.9	69.4	73.3	72.1	64.9	51.4	38.3	19.6	(20)	
	2%	DB	15.0	27.3	46.2	66.1	80.8	87.9	88.5	84.6	76.8	61.0	40.2	17.4	(21)	
		MCWB	12.2	22.8	35.8	47.6	58.7	67.6	72.4	70.9	62.7	49.2	33.5	15.1	(22)	
	5%	DB	11.5	21.9	40.9	60.2	75.4	84.0	85.3	81.9	72.8	55.7	35.1	13.0	(23)	
		MCWB	9.3	18.2	32.3	44.7	55.8	66.5	71.1	69.2	60.7	45.4	30.2	10.8	(24)	
	10%	DB	8.1	17.3	36.5	55.5	70.5	80.2	82.4	78.8	68.5	51.5	30.9	9.2	(25)	
		MCWB	6.2	14.4	29.8	42.3	53.9	65.4	70.0	68.1	58.0	42.9	27.4	7.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	27.7	39.7	53.3	62.9	74.1	77.2	75.6	68.1	55.3	38.5	20.2	(27)	
		MCDB	19.2	33.0	50.8	72.3	78.3	86.4	86.4	83.8	76.0	63.9	46.1	22.5	(28)	
	2%	WB	12.4	22.9	36.1	48.6	60.4	70.9	75.0	73.2	65.0	50.1	33.8	15.1	(29)	
		MCDB	15.0	26.9	44.8	63.3	76.0	82.6	83.4	80.2	72.9	58.4	39.5	17.3	(30)	
	5%	WB	9.3	18.3	32.9	45.7	58.0	68.8	73.4	71.2	62.6	46.8	30.7	10.8	(31)	
		MCDB	11.5	21.7	39.9	58.2	71.7	79.6	81.5	78.7	70.2	54.3	34.8	12.8	(32)	
	10%	WB	6.3	14.4	30.2	43.0	55.9	67.0	71.9	69.5	60.1	43.7	27.2	7.5	(33)	
		MCDB	8.1	17.2	36.2	54.1	67.6	77.2	79.6	76.2	66.3	50.8	30.8	9.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	20.4	19.3	19.9	20.6	18.9	16.3	16.5	19.0	16.7	14.9	15.0	(35)	
		MCDBR	19.2	23.9	22.6	29.4	29.7	25.5	21.3	21.2	25.3	24.4	19.1	16.9	(36)	
	5% WB	MCWBR	17.0	19.8	14.6	15.9	12.3	10.2	8.7	9.4	12.3	14.0	13.4	14.7	(37)	
		MCWBR	19.0	23.9	21.1	26.7	25.2	21.2	17.4	17.5	20.4	22.1	18.3	16.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.272	0.302	0.374	0.508	0.567	0.501	0.483	0.430	0.393	0.386	0.319	0.278	(40)	
		taud	2.161	2.090	1.954	1.788	1.731	2.005	2.109	2.242	2.264	2.118	2.150	2.118	(41)	
	Ebn at Noon		238	259	256	232	224	244	246	254	251	222	213	212	(42)	
		Edh at Noon	24	34	47	63	69	53	47	40	35	33	24	22	(43)	
(44) All-Sky Solar Radiation	RadAvg		531	913	1400	1586	1725	1854	1718	1526	1226	816	525	412	(44)	
	RadStd		17	31	89	168	142	236	144	107	103	49	30	23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page